Group Homework 1:- Proxy Mismatch and Optimization

Olabode Ajayi, Meredith Newman, Thabani Shumba, Jack Chung, Yusuf Ozdemir Arslan

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1 Declaration on the Use of Generative AI

In this course, we use generative artificial intelligence (GenAI) tools such as Claude and ChatGPT to better understand the concept of loss function (proxy objective) and utility function (intended human objective) in the context of Responsible ML and Alignment Problems. We declare that this was done with integrity and transparency, and that some aspects of GenAI were used as learning aids during brainstorming, preparatory research, and exploratory analysis. Therefore, the final report is a product of our own understanding and synthesis of the concepts, and we have not directly copied any content generated by GenAI. We have cited any sources or references that informed our work, and we have ensured that the report reflects our own critical thinking and insights on the topic.

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2 Problem Given

Let

$$h_L^* = \arg\min_{h\in\mathcal{H}} L(h), \quad h_U^* = \arg\max_{h\in\mathcal{H}} U(h)$$

(a) State a formal condition under which

$$\arg\min_{h\in\mathcal{H}} L(h) = \arg\max_{h\in\mathcal{H}} U(h)$$

(b) Construct an explicit counterexample: define a simple hypothesis space $\mathcal{H}$ and explicit functions $L(h)$ and $U(h)$ for which the argmin and argmax differ.

(c) Interpret your example in a real-world ML setting (e.g., credit, hiring, risk scoring): identify the proxy objective and the intended human objective.

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3 The Solutions

The Solution to the problem is structured into three parts corresponding to the sub-questions (a), (b), and (c).

3.1 (a) Formal Condition and Mathematical Explanation

With reference to the problem statement given above, we let $L(h)$ be a loss function (proxy objective) and $U(h)$ be a utility function (intended human objective) defined over a hypothesis space $\mathcal{H}$.

We seek a condition under which:

$$\arg\min_{h\in\mathcal{H}} L(h) = \arg\max_{h\in\mathcal{H}} U(h)$$

By Mathematical Explanation:

We notice that this equality holds if the minimizer of $L(h)$ is also the maximizer of $U(h)$. This is true if $U(h)$ is a strictly decreasing function of $L(h)$, That is.,

$$U(h) = f(L(h)), \quad \text{where } f \text{ is strictly decreasing}$$

For example, $U(h) = -L(h)$ or $U(h) = a - bL(h)$ for $b > 0$.

Using the Order Preservation:

For all $h_1, h_2 \in \mathcal{H}$,

$$L(h_1) < L(h_2) \implies U(h_1) > U(h_2)$$

This means the ordering of hypotheses by $L(h)$ is reversed by $U(h)$, so the hypothesis with the lowest loss has the highest utility. For instance, if $L(h)$ is the error rate of a classifier and $U(h)$ is the negative error rate (i.e., accuracy), then minimizing $L(h)$ is equivalent to maximizing $U(h)$. Therefore, the condition for the argmin and argmax to happen at the same time is that $U(h)$ is a strictly decreasing function of $L(h)$, ensuring that the optimal hypothesis under the proxy objective is also optimal under the intended human objective.

3.2 (b) Mathematical Counterexample and Explanation

Let $\mathcal{H} = \{h_1, h_2, h_3\}$.

We define the proxy objective $L(h)$ and the intended human objective $U(h)$ as follows:

- $L(h_1) = 1$, $L(h_2) = 2$, $L(h_3) = 3$

- $U(h_1) = 2$, $U(h_2) = 5$, $U(h_3) = 1$

To find the argmin and argmax:

We compute:

- $\arg\min_{h \in \mathcal{H}} L(h) = h_1$ (since $L(h_1)$ is smallest)
- $\arg\max_{h \in \mathcal{H}} U(h) = h_2$ (since $U(h_2)$ is largest)

By Mathematical Reason:

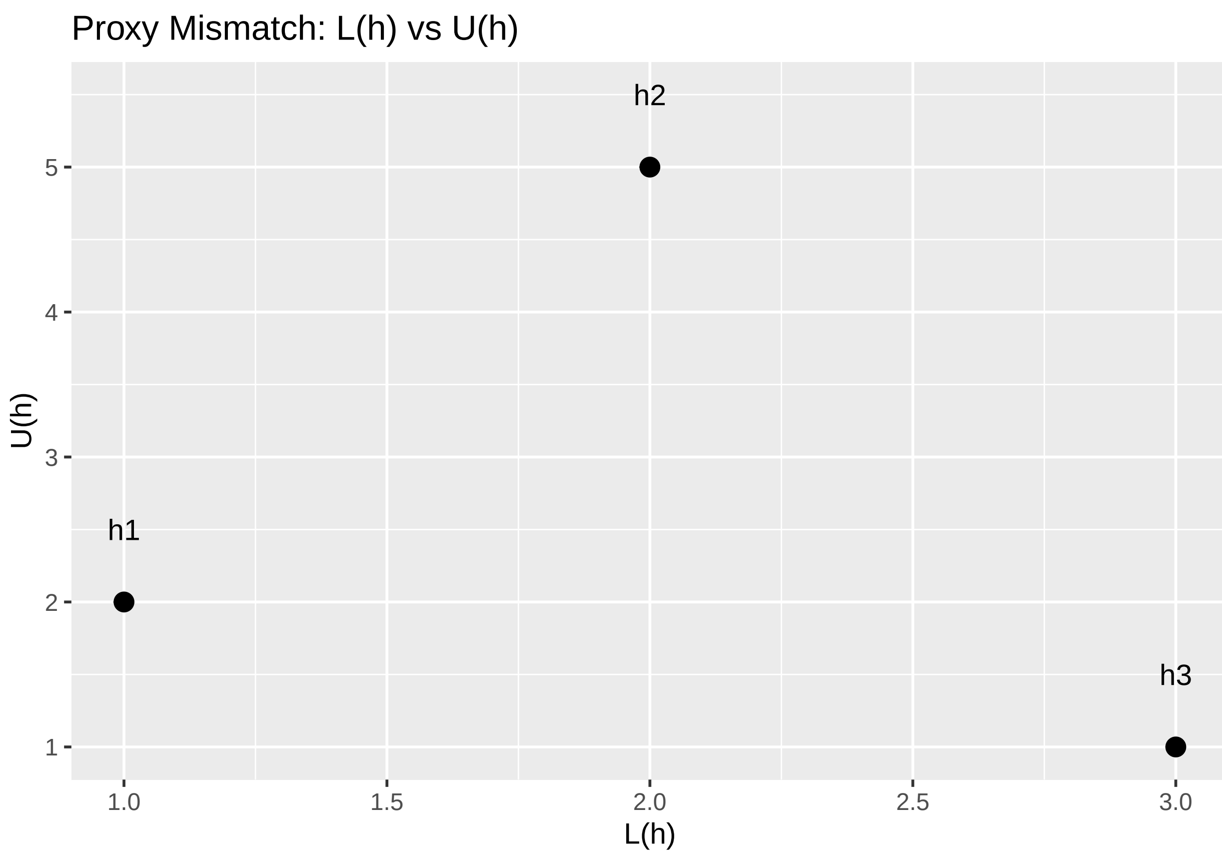
Here, $U(h)$ is not a strictly decreasing function of $L(h)$. For instance, $U(h_2)$ is highest, but $L(h_2)$ is not lowest. The mapping between $L(h)$ and $U(h)$ is not monotonic and does not satisfy the condition from part (a). Therefore, the hypothesis that minimizes $L(h)$ is not the same as the one that maximizes $U(h)$, illustrating a proxy mismatch.

With General Explanation:

If $U(h)$ and $L(h)$ are not perfectly (negatively) correlated, optimizing the proxy objective $L(h)$ may not optimize the intended objective $U(h)$. This is a classic case of proxy mismatch and can lead to suboptimal outcomes if the learning system is only optimizing for $L(h)$ without considering $U(h)$.

By Visualization:

We visualize this counterexample with a scatter plot of $L(h)$ vs $U(h)$ for the three hypotheses:



We see that h_1 has the lowest $L(h)$ but not the highest $U(h)$, while h_2 has the highest $U(h)$ but not the lowest $L(h)$. This illustrates the misalignment between the proxy and human objectives.

3.3 (c) Real-World ML Interpretation

3.3.1 Our Real-World Example is Credit Scoring:

We consider a credit scoring model used by a bank to evaluate loan applications.

Assuming we have two objectives:

- **Proxy Objective ($L(h)$):** Minimize predicted loan default rate (what the model is trained to optimize).
- **Human Objective ($U(h)$):** Maximize financial inclusion or customer satisfaction (what society or stakeholders actually care about).

Suppose we have three candidate models:

- h_1: Model that predicts lowest default rate but excludes many applicants.
- h_2: Model that predicts moderate default rate but includes more applicants and increases satisfaction.
- h_3: Model that predicts highest default rate but maximizes inclusion.

We let:

- $L(h_1) = 1$, $L(h_2) = 2$, $L(h_3) = 3$
- $U(h_1) = 2$, $U(h_2) = 5$, $U(h_3) = 1$

Interpretation:

We see that

- The model trained to minimize $L(h)$ (default rate) selects h_1.
- The true human goal (maximizing inclusion/satisfaction) would select h_2.

Proxy Mismatch:

We see that the proxy (default rate) does not align with the intended human objective (inclusion/satisfaction), leading to different choices. Hence, optimizing for the proxy can lead to suboptimal outcomes in terms of the true human objective. This is related to the broader issue of responsible AI, where we need to ensure that the objectives we optimize for in our models align with the values and goals of the stakeholders.

3.3.2 Another Example is the Hiring Algorithm

Assuming we have a hiring algorithm that evaluates candidates based on a proxy objective (e.g., test scores) while the intended human objective is to maximize employee performance and satisfaction.

We assumed the following:

- **Proxy Objective ($L(h)$):** Minimize time-to-hire or maximize test scores.
- **Human Objective ($U(h)$):** Maximize employee performance or satisfaction.

Suppose we have three candidates:

- h_1: Candidate with highest test score, but poor teamwork skills.
- h_2: Candidate with moderate test score, but excellent teamwork skills.

- h_3: Candidate with lowest test score, but high creativity.

Let:

- $L(h_1) = 1$, $L(h_2) = 2$, $L(h_3) = 3$
- $U(h_1) = 2$, $U(h_2) = 5$, $U(h_3) = 1$

Interpretation:

The algorithm picks h_1 (lowest L), but the best employee is h_2 (highest U). Therefore, optimizing for the proxy (test scores) does not lead to the best outcome in terms of employee performance and satisfaction, illustrating a proxy mismatch, and satifying the condition in part (b).

Summary Table:

Table 1: Proxy vs Human Objective Selection

Hypothesis	Proxy.Objective.L.h.	Human.Objective.U.h.	Selected.by.Proxy	Selected.by.Human
h1	1	2		
h2	2	5		
h3	3	1		

The table above summarizes the proxy and human objectives for each hypothesis, showing the mismatch in selection based on the different objectives.

In Conclusion (Our Take Away):

When the proxy and human objectives are misaligned, optimizing the proxy can lead to choices that do not maximize the intended outcome. This is a fundamental challenge in responsible AI and ML system design, and these can leads to unintended consequences if not carefully addressed. It is crucial to ensure that the objectives we optimize for in our models are well-aligned with the values and goals of the stakeholders to avoid proxy mismatch and its associated risks.

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4 Appendix

4.1 Definitions of Symbols and Parameters

Symbol	**Meaning**
$\mathcal{H}$	The hypothesis space: the set of all candidate models or decision rules under consideration.
h	A single hypothesis or model in $\mathcal{H}$.
$L(h)$	The proxy loss or optimization objective used by the learning system. The model training procedure minimizes this quantity.
$U(h)$	The true human utility or intended objective. This represents what the human decision-maker actually cares about and wants to maximize.

Symbol	Meaning
h_L^*	The hypothesis that is optimal under the proxy objective, i.e., the minimizer of $L(h)$.
h_U^*	The hypothesis that is optimal under the intended human objective, i.e., the maximizer of $U(h)$.

4.2 Example of Argmin and Argmax (from Section (b))

- $\arg\min_{h\in\mathcal{H}} L(h) = h_1$ (the hypothesis that minimizes the proxy loss)
- $\arg\max_{h\in\mathcal{H}} U(h) = h_2$ (the hypothesis that maximizes the true utility)

Explanation:

In the counterexample from Section (b), h_1 is the hypothesis that achieves the lowest value of the proxy loss $L(h)$, so it is selected by the learning system if we optimize only the proxy. However, h_2 achieves the highest value of the true human utility $U(h)$, so it would be preferred if we could directly optimize for the intended objective. This illustrates a proxy mismatch: the hypothesis chosen by minimizing $L(h)$ is not the same as the one chosen by maximizing $U(h)$.